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Real-time detection and monitoring of public littering behavior using deep learning for a sustainable environment

Eaman Alharbi^{1,2}✉, Ghadah Alsulami¹, Sarah Aljohani¹, Waad Alharbi¹ & Somayah Albaradei^{1,2}

With the global population surpassing 8 billion, waste production has skyrocketed, leading to increased pollution that adversely affects both terrestrial and marine ecosystems. Public littering, a significant contributor to this pollution, poses severe threats to marine life due to plastic debris, which can inflict substantial ecological harm. Additionally, this pollution jeopardizes human health through contaminated food and water sources. Given the annual global plastic consumption of approximately 475 million tons and the pervasive issue of public littering, addressing this challenge has become critically urgent. The Surveillance and Waste Notification (SAWN) system presents an innovative solution to combat public littering. Leveraging surveillance cameras and advanced computer vision technology, SAWN aims to identify and reduce instances of littering. Our study explores the use of the MoViNet video classification model to detect littering activities by vehicles and pedestrians, alongside the YOLOv8 object detection model to identify individuals responsible through facial recognition and license plate detection. Collecting appropriate data for littering detection presented significant challenges due to its unavailability. Consequently, project members simulated real-life littering scenarios to gather the required data. This dataset was then used to train different models, including LRCN, CNN-RNN, and MoViNets. After extensive testing, MoViNets demonstrated the most promising results. Through a series of experiments, we progressively improved the model's performance, achieving accuracy rates of 93.42% in the first experiment, 95.53% in the second, and ultimately reaching 99.5% in the third experiment. To detect violators' identities, we employed YOLOv8, trained on the KSA vehicle plate dataset, achieving 99.5% accuracy. For face detection, we utilized the Haar Cascade from the OpenCV library, known for its real-time performance. Our findings will be used to further enhance littering behavior detection in future developments.

In today's world, the consumption patterns of individuals are increasing rapidly. With the global population exceeding 8 billion, consumption rates are soaring¹. However, this raises a critical question: where does all this consumption lead? While many responsible individuals dispose of their waste properly, such as placing a water bottle into a trash bin, others thoughtlessly discard it on the ground or near the sea, allowing waves to carry it away. These destructive behaviors have a detrimental impact on our environment, as numerous studies have shown. Specifically, microplastics from these bottles significantly and adversely affect marine and soil ecosystems. This underscores the importance of raising awareness about the harmful effects of microplastics to safeguard our planet and urban spaces from pollution.

The vision for future cities anticipates the existence of smart cities, which require systems to deter uncivilized and prohibited behaviors that jeopardize their integrity. Such behaviors include littering, graffiti, tampering with public utilities, and other actions detrimental to the city and its environment. The need for such systems extends beyond smart cities, advocating for their widespread implementation to mitigate pollution and enhance the quality of life for all. Littering and improper waste disposal in public places are causing a decline in quality of life and a lack of individual responsibility towards the environment. This pollution can lead to chemical seepage into soil, contaminating groundwater supplies, affecting human health, and harming ecosystems and marine life. Animals can also be entangled in waste materials, leading to injury or death. The aesthetic deterioration caused

¹Computer Science Department, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah, Saudi Arabia. ²Center of Research Excellence in Artificial Intelligence and Data Science, King Abdulaziz University, Jeddah, Saudi Arabia. ✉email: ealraddadi@kau.edu.sa

by litter and improper waste also negatively impacts the community, property values, and well-being. To mitigate these issues, it is crucial to engage individuals and hold them accountable for their actions. Implementing a system to monitor violations and hold those responsible for their actions can help.

Strategic goals for urban development include boosting the economy across various sectors and increasing employment rates². Implementing a system that detects uncivilized and prohibited behaviors, specifically littering in public places, will support these objectives, particularly as cities continue to grow and require enhanced security and cleanliness measures in entertainment, historical, and tourism sites. Currently, these responsibilities place a heavy burden on individuals managing these facilities.

There are several commercially available systems for littering detection. The WasteWatch Cam³ is a CCTV system that uses AI technology to detect littering and dumping behavior in private or public sites. It captures and reads the car's license plate number using Automatic Number Plate Recognition (ANPR), enabling authorities to identify the offender accurately and fine them. The LitterCam system⁴ is a cost-effective solution that uses IP camera technology to detect and record instances of littering. It extends the capabilities of civil enforcement officers and helps monitor highways, roadsides, or public areas prone to litter buildup. The council verifies the footage and sends it to the Driver and Vehicle Licensing Authority (DVLA) to identify the registered keeper. LitterCam also helps maintenance crews efficiently schedule clean-up efforts by detecting litter density in different locations. The iDefigo company⁵ provides a Smart Camera solution to address fly-tipping, also known as illegal dumping. It offers high-grade cameras that can be easily installed at a low cost, capturing events and transmitting images to any device. The cameras also send alerts when a trigger event, such as fly-tipping, occurs. Our system aspires to assist government authorities and facility managers, regardless of the type of tourism, historical, or entertainment venues they oversee. This includes raising awareness in society about the critical importance of maintaining a safe and clean environment by detecting uncivilized or prohibited behaviors. This initiative seeks to foster social responsibility and consideration, particularly towards the dedicated cleaners who invest long hours working outdoors. Ultimately, this endeavor aims to elevate the quality of life for citizens and residents, reduce government expenses in this field, address global pollution crises, protect the environment, and achieve numerous positive outcomes.

In this study, we introduce the SAWN (Surveillance and Waste Notification) system to address public littering through a novel integration of real-time video classification and object detection. The major contributions are:

- We combine the MoViNet model for video classification with YOLOv8 for real-time object and license plate detection, achieving superior accuracy in both tasks.
- In the absence of relevant public datasets, we created a unique dataset by simulating real-life littering activities, contributing to the scarcity of labeled littering datasets.
- The system is designed to operate efficiently in diverse environments, leveraging lightweight models for edge computing.
- Through iterative training, our models achieve 99.5% accuracy in detecting littering and 99.5% in license plate recognition, setting new benchmarks for accuracy.

While previous studies have employed action recognition or object detection independently for environmental monitoring, our approach combines both techniques in a single framework tailored for real-time applications. The seamless integration of MoViNet and YOLOv8 ensures low latency and high accuracy, which is crucial for real-time enforcement of public behavior standards.

Related work

A comprehensive summary of the research papers relevant to our project highlights both established theories and recent advancements in detecting dumping and littering activities. This detailed review helps position our work within the existing framework. Initially, we explored research studies that use technology in various fields to gain a deeper understanding of its applications. One insightful area was cheating detection. The research paper⁶ delves into cheating detection beyond garbage littering, providing insights into object detection, which can identify instances of cheating by detecting objects such as phones, other people, and books using YOLO. The study also explores action recognition and its integration with object detection. In the following subsections, we will focus on research papers detecting littering activities by vehicle drivers and pedestrians.

Research gaps in littering behavior detection and environmental monitoring include data limitations, technology integration, real-time processing, behavioral nuances, vehicle littering detection, action recognition challenges, environmental impact studies, and user engagement. Existing datasets mainly consist of images, necessitating custom datasets for effective model training. Improved algorithms and multi-classification models are needed to capture complexities without sacrificing accuracy. Research on littering from moving vehicles is limited, and current approaches face challenges in recognizing littering actions. Addressing these gaps is crucial for developing effective monitoring systems and strategies. By focusing on these areas, researchers can contribute to improved environmental management practices, enhance public awareness, and ultimately foster a cleaner, more sustainable environment.

Pedestrian littering

Several research papers aim to detect illegal garbage dumping by pedestrians in public places, striving to create sustainable and pollution-free cities. Notably, Kim and Cho⁷ and Yun et al.⁸ utilize the OpenPose library to estimate the articulation points of the dumper's hands. Kim and Cho employ YOLO to detect different types of garbage bags, while Yun et al. use a background subtraction algorithm to detect undefined objects held by a person. Kim and Cho calculate the distance between the dumper's wrist and the bag using Euclidean distance, while Yun et al. estimate all joints and make decisions based on 1D Gaussian modeling and voting. Kim and

Cho's method tracks waste bags across multiple consecutive frames using the DeepSORT object tracking model, while Yun et al. use a correlation filter tracker for real-time object tracking. Both papers detect dumping behavior by monitoring the increase in distance between the hand and the held object, achieving accuracies of 97% and 71%, respectively.

Other research approaches, such as Anusuya and Sai⁹ and Korchani and Sethom¹⁰, incorporate face detection to identify individuals throwing waste and use object detection to identify the waste being held. Korchani and Sethom's study employs action recognition with background subtraction and human joint estimation to detect garbage dumping actions in video.

Additional studies propose real-time methods for detecting littering by pedestrians. For instance, Husni et al.¹¹ utilize CNN and CNN-LSTM methods to create an alarm system, achieving 97.7% accuracy with the CNN-LSTM model. Another study by Bae et al.¹² proposes a visual surveillance system using multi-task learning with pose estimation, action classification networks, object detection with Mask R-CNN, and a CNN feature fusion network, achieving 74.3% accuracy. Husni et al.¹³ aim to detect littering behavior using RNN and pose detection, successfully applying their model to a custom dataset.

The final paper in this subsection, by LitteringActivities¹⁴, describes a real-time surveillance system to monitor human littering activities, achieving successful predictions of regular activities and littering 76 times. Despite effective performance, the system faces challenges with fast movements, which need addressing in future work.

Vehicle driver littering

Two research papers, Mahankali et al.¹⁵ and Jain et al.¹⁶, focus on identifying and addressing illegal garbage dumping from moving vehicles through video analysis. Both studies utilize vehicle frame detection using the Haar-cascade classifier method and background subtraction to identify vehicles. Mahankali et al. incorporate the OpenALPR library for license plate recognition, detecting garbage disposal using blob detection and background subtraction. Jain et al. use background subtraction and frame differencing to analyze the nature of the curve of foreign particles for garbage detection. Both papers created custom datasets for training their systems.

Mahankali et al. report promising performance with fewer false positives as the dataset size increases, achieving 82% accuracy and 86% precision. Jain et al. achieve a high recognition rate with 93.7% accuracy. Both papers propose future improvements, such as increasing the dataset size, retraining modules, and applying the models to larger sets of video clips from roadside CCTV footage.

An additional study by¹⁷ aims to detect littering behavior linked to the license plate number to identify individuals responsible, achieving 62% accuracy in littering detection and 91% for license plate recognition. The final research by Wan Ismail et al.¹⁸ describes a system using sensors and a Raspberry Pi camera to detect illegal dumping from vehicles in Malaysia, achieving 78.08% to 88.13% accuracy in plate number recognition and providing real-time notifications to authorities.

Dataset

A robust machine learning model relies fundamentally on a well-constructed dataset. This dataset comprises a structured collection of data points or examples that the model utilizes to learn patterns, make predictions, and perform various tasks. Our objective is to develop an efficient real-time littering action detection model. To achieve this, we actively searched for diverse online datasets. However, we discovered that all available datasets consist solely of images, none depicting littering actions or identifying violators. Consequently, we simulated littering scenarios and collected our own dataset. Additionally, we gathered Saudi Arabian license plates available online to facilitate vehicle license plate detection in our simulated dataset. For facial identity detection, we leveraged the OpenCV library, negating the need to collect a separate dataset.

Our dataset consists of 276 videos, which were split into 70% (193 videos) for training, 15% (41 videos) for validation, and 15% (42 videos) for testing. This split follows best practices to ensure the model is trained, validated, and tested on distinct datasets, with the test dataset kept free from any augmentation to maintain unbiased evaluation. Since the videos were captured using various cameras, we standardized their dimensions and quality by resizing all videos to a 16:9 aspect ratio and 1080p HD resolution. Additionally, the audio was muted across all videos to avoid unnecessary processing overhead. After these preprocessing steps, the videos were prepared for the augmentation phase, applied only to the training and validation datasets to enhance the model's generalization ability.

Augmentation

Our dataset was divided into 80% for training and 20% for testing. To enhance the dataset, we implemented various augmentation techniques such as vertical flipping, horizontal flipping, multiplication, and Gaussian blur. These techniques were applied to each video, sometimes combining more than one technique to increase variability and robustness. For a visual representation of our dataset, please refer to Fig. 1.

Data augmentation includes automated procedures performing geometric and color transformations on images. We evaluated the benefits of flipping, brightening, and darkening. Flipping methods included horizontal, vertical, and combined flips, each generating new images from the originals. Brightening and darkening transformations were applied to create images with varying brightness levels.

Proposed methodology

The SAWN system has three layers: the AI model, the front end, and the back end. Integration of these layers requires careful execution of project development phases.

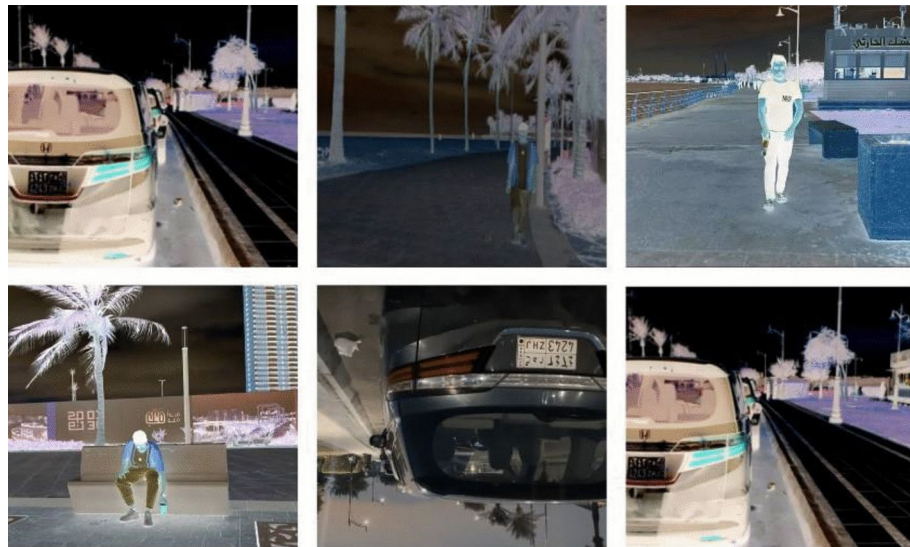


Fig. 1. An example of the augmentation that we did on our dataset.



Fig. 2. The proposed methodology in brief.

- **AI model development:** The most important stage of developing our system which involves recognizing littering behavior through video classification techniques.
- **Front-end development:** the process of designing and implementing the visible components of web applications. It includes creating the user interface and user experience that users directly interact with. This aspect of web development is responsible for how a website looks, feels, and behaves in a web browser.
- **Back-end development:** It deals with the server-side of web applications. It involves server-side scripting, databases, and server configurations that work behind the scenes to support the front-end of websites. The implementation of the AI model consists of four phases, as illustrated in Fig. 2. These phases involve detecting

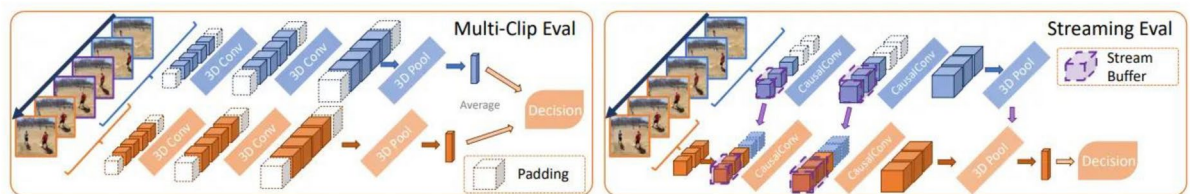


Fig. 3. MoViNet architecture (Source Kondratyuk et al.¹⁹).

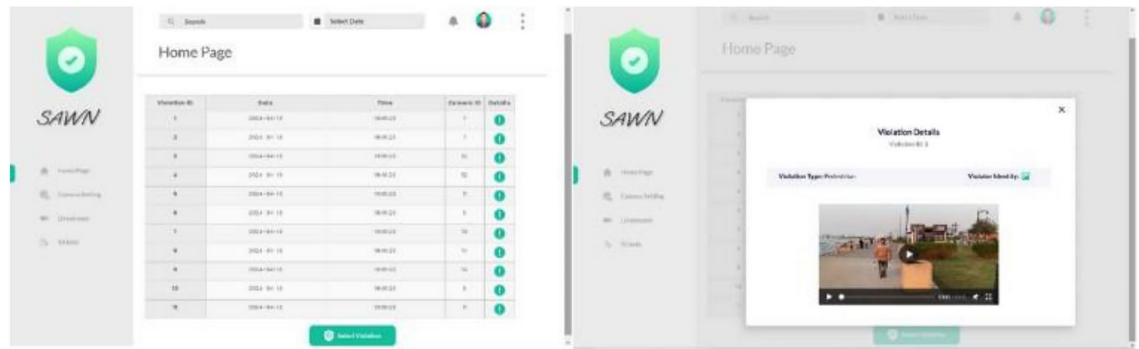


Fig. 4. This figure explains littering detection with its behavior.

littering violations using an action detection model, identifying violators using object detection models, storing the violations in the system, and transmitting the violations to an external system for fines.

1. Detect Littering Behavior from surveillance cameras: Surveillance camera footage is used to identify instances of littering. The footage is analyzed using a video classification technique to predict whether the littering behavior originates from a pedestrian or a vehicle. The AI model used for detecting littering behaviors is the Mobile Video Networks (MoViNets) model. MoViNets are a convolutional neural network architecture that Google researchers built to be efficient in computation and memory usage, making them highly effective for real-time streaming applications Fig. 3. MoViNet stands out from other video classification networks by addressing scalability and inference speed issues. It uses a diverse set of efficient 3D CNN architectures for mobile platforms, uncoupling memory from video clip duration for minimal memory footprint. MoViNets also introduce a model ensemble technique, training two models independently with half the original video frame per second, resulting in stable results while maintaining the same computational complexity and model size. They represent a significant improvement over the 3D CNN model, which, despite its accuracy, requires substantial computational power and memory, making it unsuitable for mobile devices. The model was trained using a specifically annotated (SAWN) dataset, which includes videos categorized into two distinct groups: pedestrian littering and vehicle littering.
2. Detect Violator's Identity: To address violations, it is important to identify the individuals responsible. For face detection violations, OpenCV provides real-time libraries that are effective in recognizing faces, and one of them is the Haar Cascades library, which has proven to be efficient in detecting faces in real-time footage²⁰. The Haar Cascade model from the OpenCV library, a machine learning-based method for detecting faces, was employed. This classifier has been found to be highly effective for face detection²¹. The model uses a cascade function trained on numerous images and is then applied to the detected littering violation video. This process analyzes one frame at a time without requiring additional training on the dataset. For real-time, general-purpose face detection, we selected a pre-trained classifier due to its balance between accuracy and speed.

To identify the license plates of vehicles whose drivers were caught littering, we utilized the advanced YOLOv8 object detection model. YOLOv8, the latest version of the You Only Look Once (YOLO) algorithm, uses deep neural networks to detect objects quickly and accurately in images or video frames. YOLOv8 is designed to improve upon its predecessors by enhancing both the accuracy and speed of object detection. Using this powerful tool, we were able to identify violators and take action against those who were littering from their vehicles. The YOLOv8 model was trained using the KSA license plate dataset.

3. Store the violations on SAWN system: After preparing the models to detect violations and identify violators, we integrated them with a website to store the data in an SQLite3 database and display it. The Fig. 4 illustrates the detected violations and the associated identities.

The Flask API is a Python-based tool that connects the front end of a SAWN system to the database, model, and webcam. It supports popular AI libraries like TensorFlow, making it an ideal choice for a SAWN system. To integrate the webcam with the website, call the function Videocapture on the Livestream page. The AI model integration is triggered by two functions: running the models and communicating data flow between the model and the website. The model integration involves detecting littering behavior and checking for new violations. After the AI models are integrated, the results are stored in different folders connected to the website's backend. These folders are loaded on the website interfaces and shown to the admin user. The 'check for new violations()' function updates the backend of the website and displays it on the home page after adding new violations data to the folders (Fig. 5).

4. Send the violations to the external entity: After logging violations on the website, the administrator can transmit this information to external entities to levy fines or initiate required measures. The subsequent Fig. 6 outlines the process for sending this information to an external entity.

Experimental analysis and results

1. MoViNet: The model was initially trained on the SAWN dataset using a multi-classification technique with four classes:

Pedestrian Littering, Vehicle Littering, Pedestrian Normal, and Vehicle Normal. The training accuracy was 95.52%, and the testing accuracy was 78.95% (Fig. 7). These results suggest that the model may have over-fitted the training set. To address this, the model was trained on different dataset arrangements, excluding confusing class pairs such as Pedestrian Littering and Pedestrian Normal, as well as Vehicle Littering and Vehicle Normal, which were causing confusion for the model. The best results were achieved by training the model on a binary classification with two classes: Pedestrian Littering and Vehicle Littering. This experiment showed 99.5% accuracy on both the training and testing sets. Additionally, the model was evaluated on multiple performance metrics, achieving Precision: 98.9%, Recall: 97.8%, and an F1-Score: 98.35%, further demonstrating its robustness and reliability.

2. YOLOv8: The YOLOv8 model underwent training using the KSA license plate dataset. Figure 8 presents the confusion matrix, which shows the results of analyzing vehicle license plates. The matrix illustrates the instances where the model correctly predicted the license plate class. This model has been trained with an accuracy of 99.5%. Additionally, Fig. 9 shows snapshots taken from real-time footage, where the model accurately detected the plates with 90% accuracy. The image in Fig. 10 displays the cropped region of interest of the license plate.
3. Haarccascade: In this particular step of face detection, the process of training the model on the dataset is not required. Instead, we made use of the Haar cascade pre-trained model to predict the outcome. It is worth mentioning that the Haar cascade has already been trained on a large dataset, which enables it to recognize specific features and patterns. The Haar cascade classifier was used in this experiment to detect faces present in the video. Once detected, a bounding box was drawn around the face on the frame, and the identified face was cropped and displayed. The experiment was successful in detecting faces within the footage, and it pre-

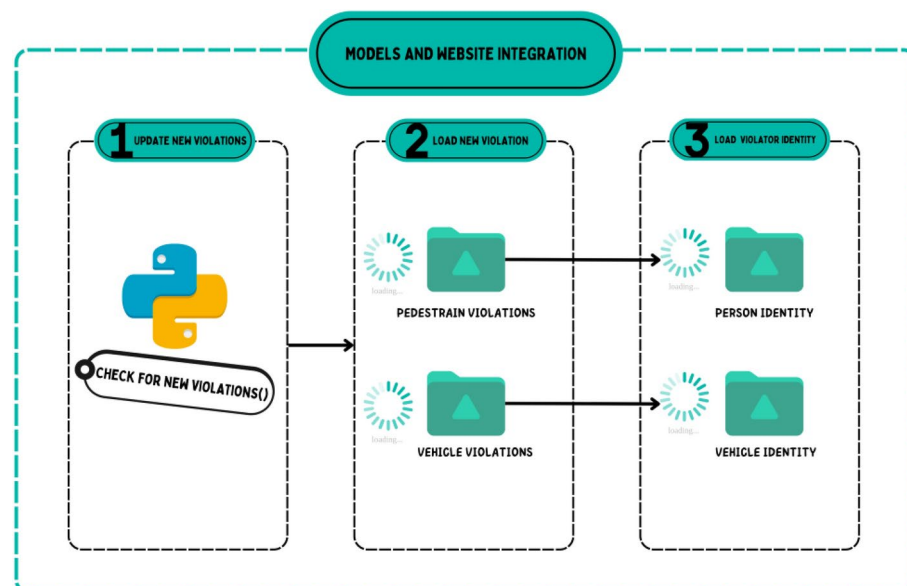


Fig. 5. Model and website integration.

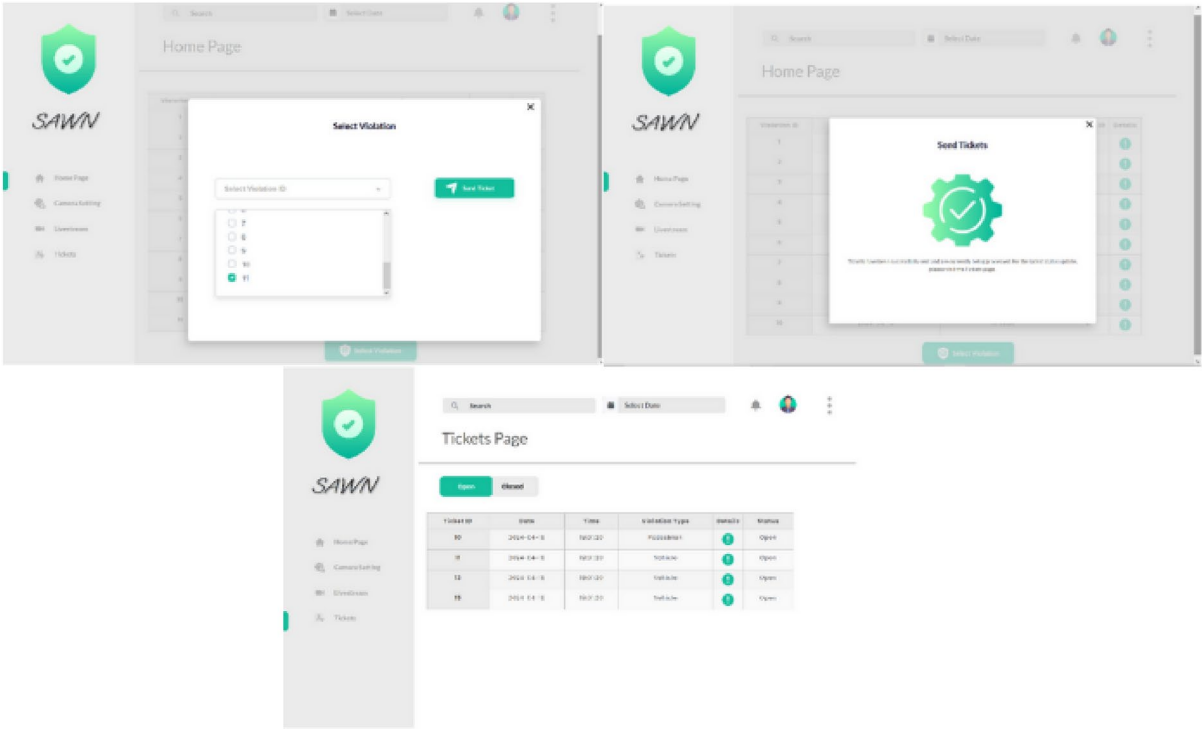


Fig. 6. This figure describes the sent littering process.

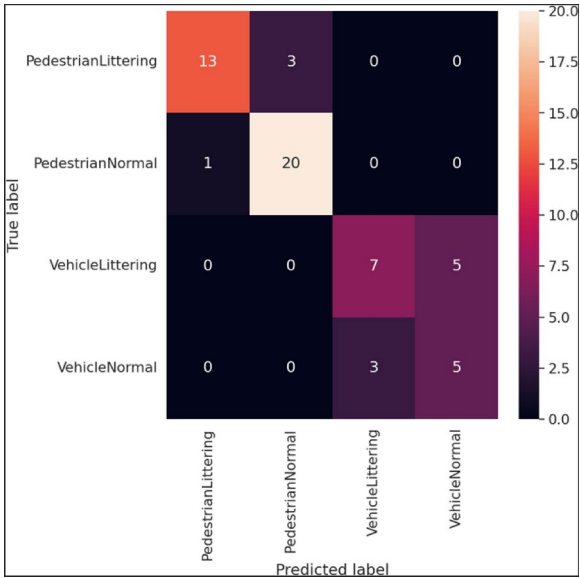


Fig. 7. The confusion matrix of the multi classification model.

cisely identified facial features, which contributed to our objectives in tracking littering violations. Overall, using a pre-trained model can save time and be an effective way to achieve accurate results without requiring extensive training.

To demonstrate the novelty and effectiveness of our approach, we compared the performance of our model with recent frameworks in the field of littering detection and behavior monitoring. Table 1 summarizes the comparison of our results with other state-of-the-art models based on key performance metrics such as accuracy, precision, recall, and F1-score.

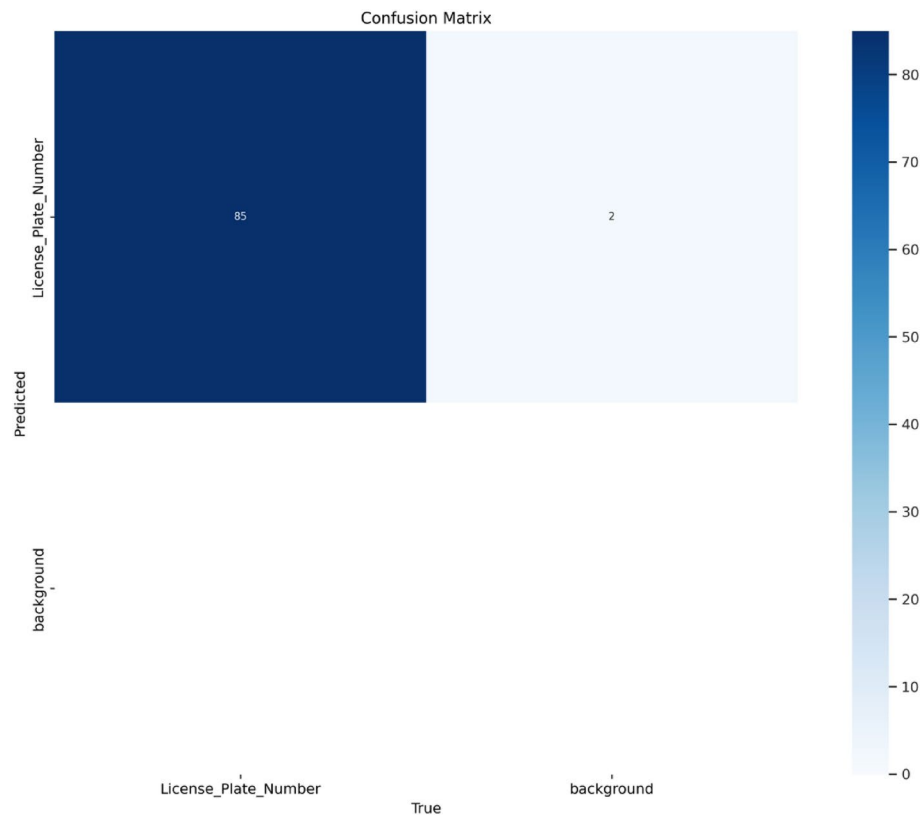


Fig. 8. The confusion matrix of the vehicle plate model.



Fig. 9. A video of vehicle littering incident.



Fig. 10. The identified vehicle plate image output.

Study	Model used	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Kim and Cho ⁴	YOLO + OpenPose	97.0	96.5	95.2	95.85
Mahankali et al. ¹²	Haar cascade + ALPR	93.7	92.0	90.5	91.2
Bae et al. ⁹	Mask R-CNN + CNN fusion	74.3	70.8	72.5	71.65
This study	MoViNet + YOLOv8	99.5	98.9	97.8	98.35

Table 1. Comparison with state-of-the-art (SOTA) models.

Discussion

The results of our experiments indicate that MoViNet is highly effective in detecting littering behaviors with an accuracy of 99.5%. However, the initial multi-classification approach revealed limitations, particularly in distinguishing between similar behaviors such as pedestrian littering and normal pedestrian activity, as well as vehicle littering and normal vehicle activity. By simplifying the classification to a binary model, we achieved 99.5% accuracy, highlighting the potential of MoViNet for targeted applications. Nonetheless, this binary classification approach may overlook nuanced behaviors that a more sophisticated multi-class model could capture. Therefore, future work should focus on refining the multi-classification model to better handle these nuances without compromising accuracy.

Our work builds upon previous studies that utilized various machine learning and computer vision techniques to detect littering and other environmental violations. For instance, Kim and Cho⁷ employed YOLO and OpenPose for detecting illegal garbage dumping by pedestrians, achieving an accuracy of 97%. Similarly, Mahankali et al.¹⁵ used a combination of Haar-cascade classifiers and background subtraction for detecting littering from moving vehicles, reporting an accuracy of 93.7%. While these studies demonstrated effective detection methods, our approach with MoViNet and YOLOv8 presents a notable improvement in accuracy and efficiency. By leveraging the computational efficiency of MoViNet and the robust object detection capabilities of YOLOv8, we achieved higher accuracy rates and more reliable real-time performance. This comparison underscores the advancements our system brings to the field, particularly in the context of real-time streaming applications and multi-environment adaptability.

Furthermore, the integration of YOLOv8 and Haar Cascade models significantly enhanced our ability to identify violators. YOLOv8's high accuracy in detecting vehicle license plates (99.5%) demonstrates its robustness and reliability for real-time applications. Similarly, the Haar Cascade model effectively identified pedestrian violators through facial recognition. However, the success of these models depends on the quality and variety of the training data. Future work should include expanding the dataset to cover a broader range of scenarios and conditions to ensure the models' robustness and adaptability in diverse environments.

The proposed approach to litter detection and monitoring uses MoViNet and YOLOv8, achieving an impressive 99.5% accuracy in detecting littering behavior. This is a significant improvement from previous studies, which achieved 97% accuracy using YOLO and OpenPose⁷ and Mahankali et al.¹⁵ who reported 93.7% accuracy with Haar-cascade classifiers and background subtraction. The integration of MoViNet enhances computational efficiency, allowing real-time detection in various environments. The system is designed to adapt to different environments, making it essential for effective litter detection in diverse public spaces. It combines action detection with object detection, allowing for the identification of violators through facial recognition and license plate detection. The authors acknowledge the need for refining the multi-classification model to capture more nuanced behaviors, demonstrating a commitment to continuous improvement in the rapidly evolving field of computer vision and machine learning. However, the study faced challenges in collecting appropriate data for littering detection, leading to the simulation of real-life scenarios.

The team faced several challenges during their project development journey, including scope determination, research, dataset collection, and dataset collection. Scope determination required a deep understanding of the project's scope, focusing on the most important aspects. Research was crucial in selecting the most suitable technology and methods for the project, such as video classification, which helped identify the most effective approach to detect littering behavior. Dataset collection was a challenge due to the lack of resources, but perseverance and exploration of various tools and techniques helped develop new skills and expand knowledge of working with complex data. There were some challenges in integrating models and a website, including moving models from one platform to another and encountering issues with the MoViNet model due to a new update.

While our current system addresses littering, it is important to acknowledge other forms of environmental violations that also need attention. Future iterations of this project should aim to incorporate detection capabilities for behaviors such as illegal dumping, vandalism, and other activities harmful to the environment. By expanding the scope of the detection system, we can provide a more comprehensive solution that not only targets littering but also promotes broader environmental protection and sustainability efforts.

Conclusion and future work

In this paper, we propose a solution to reduce littering in public places and enhance overall quality of life. Our approach involves using MoViNet to detect littering behavior with 99.5% accuracy. However, we aim to further improve the accuracy and prediction capabilities of the multi-classification model. To identify the violators' identities, we utilized YOLOv8 to detect vehicle license plates and Haar Cascade to identify the faces of pedestrian violators. YOLOv8 demonstrated up to 99.5% accuracy in littering detection.

While our current focus is on littering, we recognize the presence of other detrimental behaviors impacting the environment. In the future, we hope to incorporate detection mechanisms for these additional behaviors to create a more comprehensive solution for environmental protection.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request.

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Author contributions

E.A., G.A., S.A., and W.A.: Conceptualization; G.A., S.A., and W.A.: data Creation; E.A., G.A., S.A., and W.A.: methodology; E.A., G.A., S.A., and W.A.: formal analysis; E.A., G.A., S.A., and W.A.: validation; E.A., S.A.: writing—original draft; E.A., G.A., S.A., W.A. and S.A.: writing—review & editing. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to E.A.

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